Exercise Sheet 3

Generative Models and Naive Bayes

Probabilistic Machine Learning

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3.A The Generative Route

Exercise 3.A.1: Counting Parameters

A fraud classifier (two classes) uses four attributes: **payment** (card, invoice, wallet), **country** (domestic, EU, other), **new device** (yes/no) and **night** (yes/no).

1. How many free numbers does the full generative model need — the class prior and one joint $p(x \mid y)$ per class?
2. How many does naive Bayes need?
3. For p yes/no attributes and K classes, give both counts as formulas.
4. From which p on does the full joint need more than a million numbers *per class*?

Exercise 3.A.1: Counting Parameters — Solution

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1. Full model. Each class needs a table over $3 \cdot 3 \cdot 2 \cdot 2 = 36$ value combinations, of which 35 are free (they sum to 1). Two classes give 70; the prior $P(\text{fraud})$ adds one: **71**.

2. Naive Bayes. Per class one categorical per attribute: $(3-1)+(3-1)+(2-1)+(2-1) = 6$ free numbers. Two classes give 12, plus the prior: **13**.

3. Formulas. Full: $K(2^p - 1) + (K - 1)$. Naive Bayes: $Kp + (K - 1)$.

4. $2^{19} - 1 = 524{,}287$ and $2^{20} - 1 = 1{,}048{,}575$, so from $p = 20$ on. The naive count grows linearly: 20 per class.

3.B Naive Bayes by Hand

Exercise 3.B.1: Twelve Transactions

Flags: new device D, foreign IP F, night N, high amount H (1 = yes).

#	D	F	N	H	fraud?	#	D	F	N	H	fraud?
1	1	1	1	1	yes	7	1	0	0	0	no
2	1	1	0	1	yes	8	0	0	0	0	no
3	1	0	1	0	yes	9	0	1	0	0	no
4	0	1	1	1	yes	10	0	0	0	0	no
5	0	0	0	0	no	11	1	0	1	0	no
6	0	0	1	0	no	12	0	0	0	0	no

1. Estimate $P(\text{fraud})$ and $P(\text{flag} = 1 \mid \text{class})$ for all four flags by counting.
2. Compute $P(\text{fraud} \mid x)$ for $x_1 = (D, F, N, H) = (1, 0, 0, 0)$.
3. Do the same for $x_2 = (0, 0, 0, 1)$. What goes wrong, and why?
4. Repeat 2 and 3 with Laplace smoothing on the flags (keep the prior unsmoothed). Which result changes a lot, which only a little?

Exercise 3.B.1: Twelve Transactions — Solution

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1. $P(\text{fraud}) = 4/12 = 1/3$. Among the 4 frauds, each flag occurs 3 times: $3/4$ for D, F, N, H. Among the 8 honest: $D\ 2/8, F\ 1/8, N\ 2/8, H\ 0/8$.

2. Fraud: $\frac{1}{3} \cdot \frac{3}{4} \cdot \frac{1}{4} \cdot \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{256} \approx 0.0039$. Honest: $\frac{2}{3} \cdot \frac{2}{8} \cdot \frac{7}{8} \cdot \frac{6}{8} \cdot 1 = 0.1094$. $P(\text{fraud} \mid x_1) = 0.0039/(0.0039 + 0.1094) = 1/29 \approx 0.034$.

3. Fraud: $\frac{1}{3} \cdot \frac{1}{4} \cdot \frac{1}{4} \cdot \frac{1}{4} \cdot \frac{3}{4} = \frac{1}{256}$. Honest: $\ldots \cdot P(H{=}1 \mid \text{honest}) = 0$. So $P(\text{fraud} \mid x_2) = 1$: a transaction without any red flag except a high amount is **certain** fraud, because no honest high amount occurred in eight rows. One zero count vetoes the other three flags, which all speak for honest.

4. Smoothed, $(\text{count} + 1)/(n + 2)$: fraud $4/6 = 2/3$ for every flag; honest $D\ 0.3, F\ 0.2, N\ 0.3, H\ 0.1$. x_1: fraud $\frac{1}{3} \cdot \frac{2}{3} \cdot (\frac{1}{3})^3 = 0.0082$, honest $\frac{2}{3} \cdot 0.3 \cdot 0.8 \cdot 0.7 \cdot 0.9 = 0.1008$, so $P = 0.075$ (from 0.034). x_2: fraud 0.0082, honest $\frac{2}{3} \cdot 0.7 \cdot 0.8 \cdot 0.7 \cdot 0.1 = 0.0261$, so $P \approx 0.24$ (from 1). With 4 and 8 rows the prior of one pseudo-count per value is not small, so even x_1 moves — smoothing matters most where the counts are smallest.

3.C Continuous Attributes

Exercise 3.C.1: Gaussian Naive Bayes by Hand

Twelve transactions: the amount as $x = \log_{10}$ amount in € ($x = 2$ is 100 €) and the flag new device D.

	x	$D = 1$
4 frauds	2, 2, 3, 3	3 of 4
8 honest	1, 1, 1, 1, 2, 2, 2, 2	2 of 8

1. Fit a normal per class to x: mean and standard deviation (divide by n).
2. A new transaction of 316 € ($x = 2.5$), device unknown. Compute $p(x \mid \text{fraud})$, $p(x \mid \text{honest})$, their ratio and $P(\text{fraud} \mid x)$ with the prior of the table.
3. It came from a new device. Update the posterior with Gaussian naive Bayes.
4. Would the likelihood ratio change if the densities were written over the amount in € instead of $\log_{10}$ amount? Why (not)?

Exercise 3.C.1: Gaussian Naive Bayes by Hand — Solution

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1. Fraud: $\hat{\mu} = 2.5$, deviations ± 0.5, so $\hat{\sigma} = 0.5$. Honest: $\hat{\mu} = 1.5, \hat{\sigma} = 0.5$.

2. $p(2.5 \mid \text{fraud}) = \frac{1}{0.5\sqrt{2\pi}}\, e^{0} = 0.798$; honest: $z = 2$, so $p(2.5 \mid \text{honest}) = 0.798\, e^{-2} = 0.108$. Ratio $e^{2} \approx 7.39$. Prior odds $4:8$, posterior odds $7.39/2 = 3.69$, $P(\text{fraud} \mid x) = 3.69/4.69 \approx 0.79$.

3. $\text{LR}_D = (3/4)/(2/8) = 3$. Posterior odds $3.69 \cdot 3 = 11.1$, $P \approx 0.92$ — assuming amount and device independent given the class.

4. No. Over the amount $a = 10^{x}$ each density gets the factor $1/(a \ln 10)$, the same in both classes at the same a; it cancels in the ratio.

Exercise 3.C.2: A Shared Spread Gives a Threshold

Both classes of 3.C.1 have the same spread, $\sigma = 0.5$.

1. Show that with equal spreads $\log \frac{p(x|\text{fraud})}{p(x|\text{honest})}$ is linear in x, and give it for the fitted numbers.
2. With the prior of the table, above which amount does the Bayes decision rule decide fraud?
3. The same model at the real base rate $1:999$: where is the cut now?
4. Write $P(\text{fraud} \mid x)$ as $1/(1+e^{-(wx+b)})$ for the prior of the table. Which model of the next lecture has this form?

Exercise 3.C.2: A Shared Spread Gives a Threshold — Solution

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1. $\log \mathrm{LR}(x) = \frac{(x-\mu_h)^2-(x-\mu_f)^2}{2\sigma^2} = \frac{\mu_f-\mu_h}{\sigma^2}\,x - \frac{\mu_f^2-\mu_h^2}{2\sigma^2}$: the x^2 terms cancel. Here $\frac{1}{0.25}\,x - \frac{6.25-2.25}{0.5} = 4x - 8$.

2. Fraud iff $4x - 8 + \ln\frac{1}{2} > 0$, i.e. $x > (8+\ln 2)/4 = 2.17$: above $10^{2.17} \approx 150$ €.

3. $x > (8+\ln 999)/4 = (8+6.91)/4 = 3.73$: above about 5,300 €. The prior moves the cut; the model is unchanged.

4. $w = 4, b = -8-\ln 2 \approx -8.69$. A logistic function of a linear score: logistic regression (Lecture 04) fits w and b directly, without a model of x.

3.D Deciding on a Class

Exercise 3.D.1: Costs, Prior and One Threshold

A naive Bayes fraud model at the base rate $1:999$. Checking a transaction costs €5 if it is honest; a fraud let through costs €500.

1. Above which posterior should a transaction be checked? Above which likelihood ratio?
2. Three transactions have likelihood ratios 6, 84 and 1,140. Which are checked?
3. Which of the three does the Bayes decision rule (equal costs) call fraud?
4. A new market has a fraud rate of 1 in 100 and the same fraud patterns. What is the threshold on the likelihood ratio there — and must the model be refitted?

Exercise 3.D.1: Costs, Prior and One Threshold — Solution

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1. $p^* = 5/505 \approx 0.0099$. As odds, $p^*/(1-p^*) = 5/500$, so $\mathrm{LR} > \frac{5}{500} \cdot 999 \approx 10$.

2. Posteriors $6/1{,}005 = 0.6\%, 84/1{,}083 = 7.8\%$ and $1{,}140/2{,}139 = 53\%$: the last two are checked.

3. Only the last: a posterior above 50% needs $\mathrm{LR} > 999$.

4. $\mathrm{LR} > \frac{5}{500} \cdot 99 \approx 1$. No refit: only the prior changes, provided $p(x \mid y)$ is the same in the new market — a generative model swaps $P(y)$ and keeps its likelihoods.